http://bit.ly/4lpV1qN (Powerpoint Slides)

https://bit.ly/4rqy8os (PDF Slides)

Dan Tripp

Senior Digital Accessibility Engineer, Siteimprove

Jeremy Gonzales

Senior Digital Accessibility Specialist, Siteimprove

IS YOUR WEBSITE "AUTOMATED CHECKER FRIENDLY?"

DIVE IN THEN STEP BACK

Two customer stories

Customer = *user* of an automated checker

CUSTOMER STORY #1: S.P.A. AT AN ENVIRONMENTAL AGENCY

- SPA = Single-Page Application, doesn’t have traditional pages and links. Single PAGE, multiple page states.
- Customer had new site: SPA
- A crawl is…

CUSTOMER STORY #1: S.P.A. AT AN ENVIRONMENTAL AGENCY

- Crawl result: one page
  - Obviously, because it’s a single-PAGE application
  - ➔ Most content: not checked
- Help Center article
- There are time-consuming ways around this.

CUSTOMER STORY #2: CAROUSEL AT A HOSPITAL NETWORK

Recent Stories

New Pediatric Wing Opens

Our expanded children's center now offers 40 additional beds and a family lounge.

1,000th Heart Surgery Milestone

Our cardiac team celebrates a landmark achievement in patient care.

Community Health Fair Success

Over 500 residents received free screenings at our annual wellness event.

Nurse of Aw

Congratula year's w outstanding to patie

CUSTOMER STORY #2: CAROUSEL AT A HOSPITAL NETWORK

- Faded out slide indicates continuation
- Automated checker: flagged contrast error
- Automated checker: no human judgement
- Customer reported on 171 pages.

GOAL OF THIS TALK

- To give a name to this as-yet-unnamed category of bad practices that we keep seeing.

GOAL OF THIS TALK

- Now that we have a name, we can say:
  - “I’m sorry but your site isn’t
  - Automated Checker Friendly.”

GOAL OF THIS TALK

- These “automated checker *un*friendly” practices result in two types of customer frustration:

1) Bad coverage (story #1)
2) Noise (story #2)

GOAL OF THIS TALK

- Why build your site in an Automated Checker Friendly way?
  - Full disclosure: we will both benefit.
  - We (the vendor) will undoubtedly benefit. Reduced support tickets etc.

GOAL OF THIS TALK

- More importantly: you (the user of the automatic checker) will benefit.
  - Because increased coverage + less noise ➔ time savings ➔ more efficient accessibility program ➔ better accessibility

GOAL OF THIS TALK

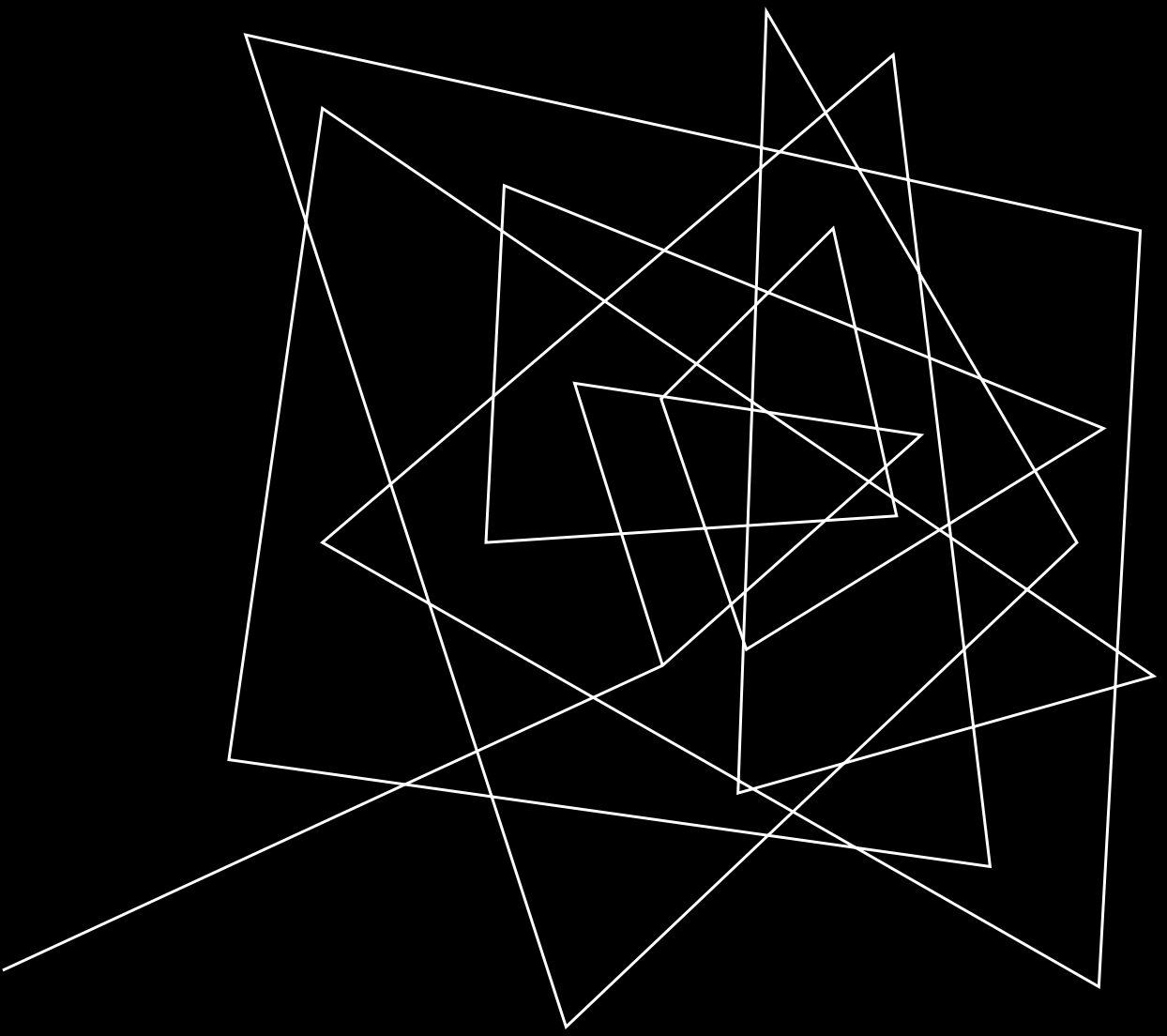
- Other situations: certainly, the automated checker needs to change.
  - Bug reports, feature requests.
- But in this talk....

ASSUMPTIONS

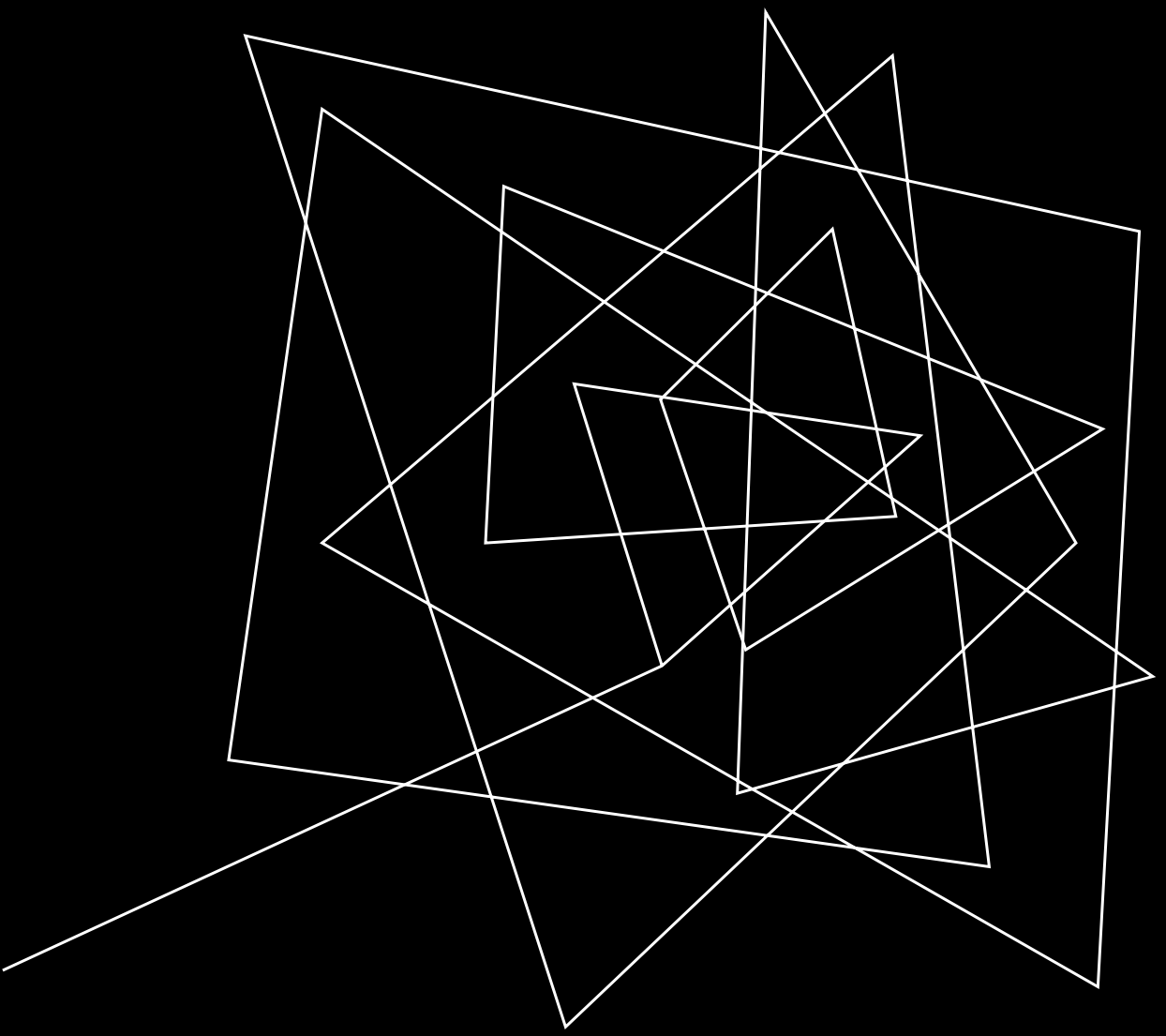
1. You either develop web pages, or you advise others on how they should make their web pages.
   a) This presentation is technical
   b) Requires understanding of web development and accessibility standards
2. Automated checkers are sometimes useful.
   a) Allows you to evaluate pages at scale
   b) Only 20-40% of WCAG coverage
   c) Comprehensive a11y strategy includes manual testing

ASSUMPTIONS

3. Conforming to WCAG is good.
   a) Standardized way of evaluating accessibility
   b) Encompasses a broad range of user types
4. *You* are good.
   a) You aren't here to learn how to cheat the automated checker
   b) You care about discovering and fixing accessibility issues to improve the user experience



FRUSTRATION TYPE #1: BAD COVERAGE



CUSTOMER STORY #3 OF 8: ACCORDIONS AT A DUTCH GOVERNMENT ORG

STORY #3 OF 8: ACCORDIONS

- Support ticket.
- Customer complaint: our automated checker wasn’t flagging accessibility problems inside their accordions.
  - Collapsed by default.
  - Problems with links.

STORY #3 OF 8: ACCORDIONS

- Average 5 lines of text per accordion.

STORY #3 OF 8: ACCORDIONS

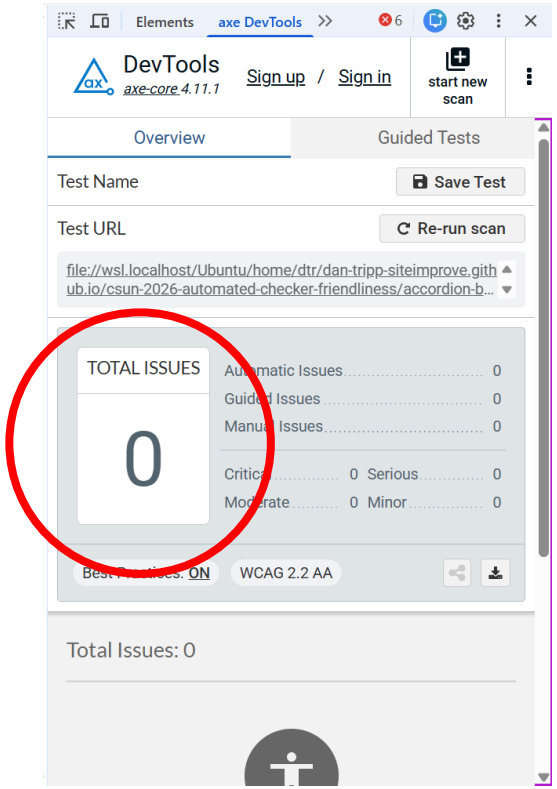
- We said: nothing we can do.
- That was the crawler. Browser extension: workaround: click around.
- Accordions are not “automated checker friendly”.
- Don't use an accordion unless you're sure you need them.
  - Alternative: headings.
  - Alternative: accordions on mobile, not desktop.
  - Alternative (time sink, not scalable): browser automation, “user flow” tools

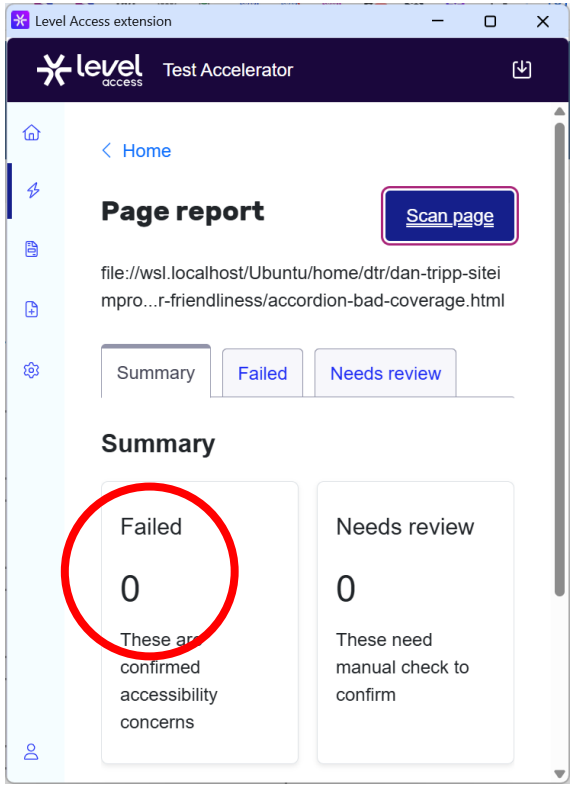
STORY #3 OF 8: ACCORDIONS

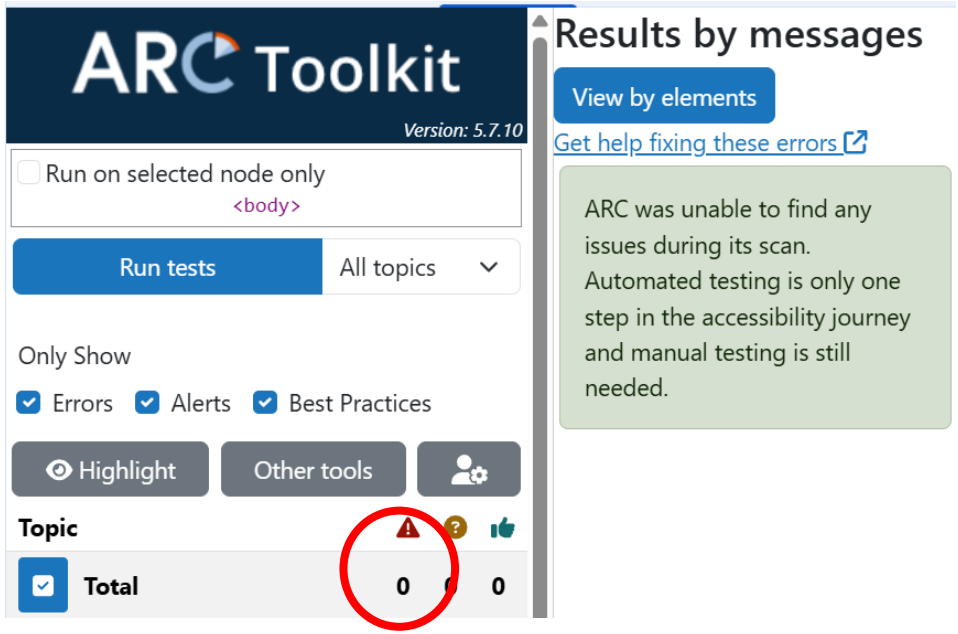
- General pattern: any "click to reveal" functionality (accordions, popups, modals, more) will result in bad coverage.

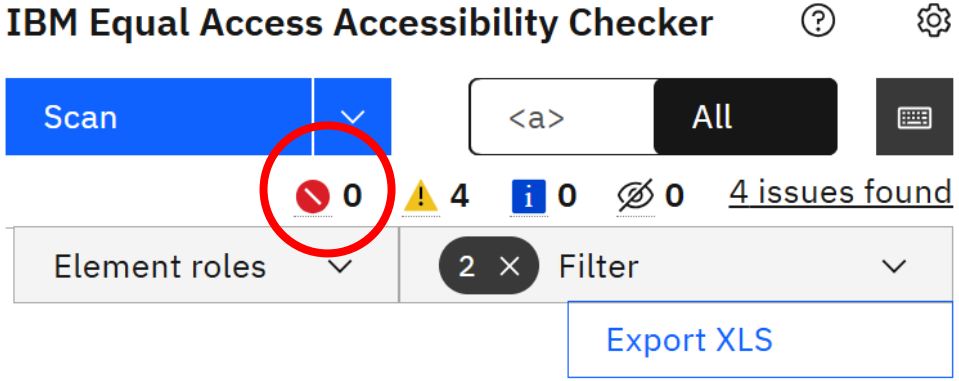
STORY #3 OF 8: ACCORDIONS

STORY #3 OF 8: ACCORDIONS

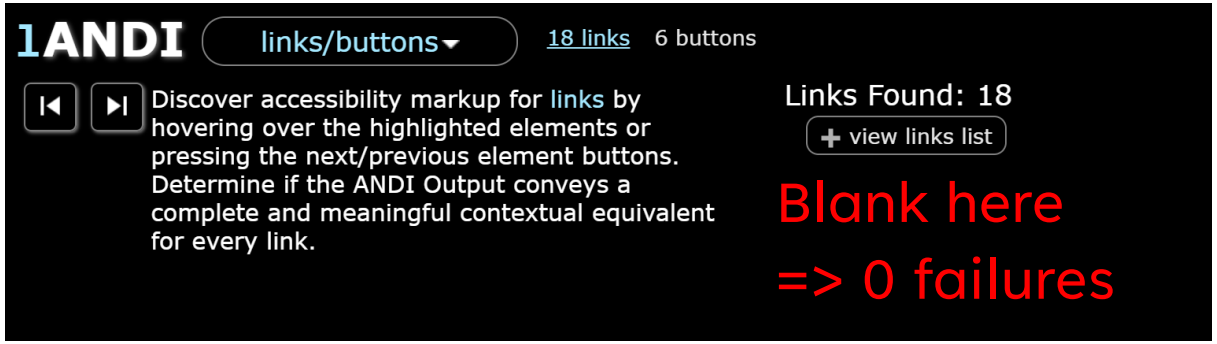






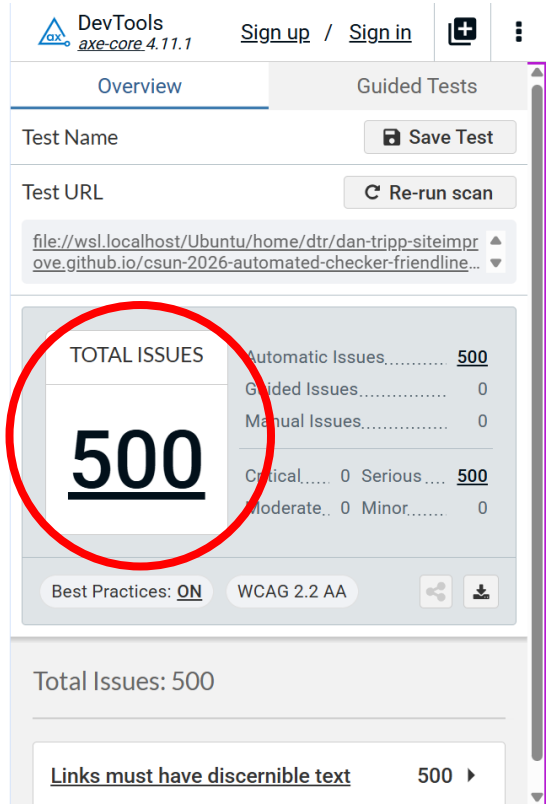


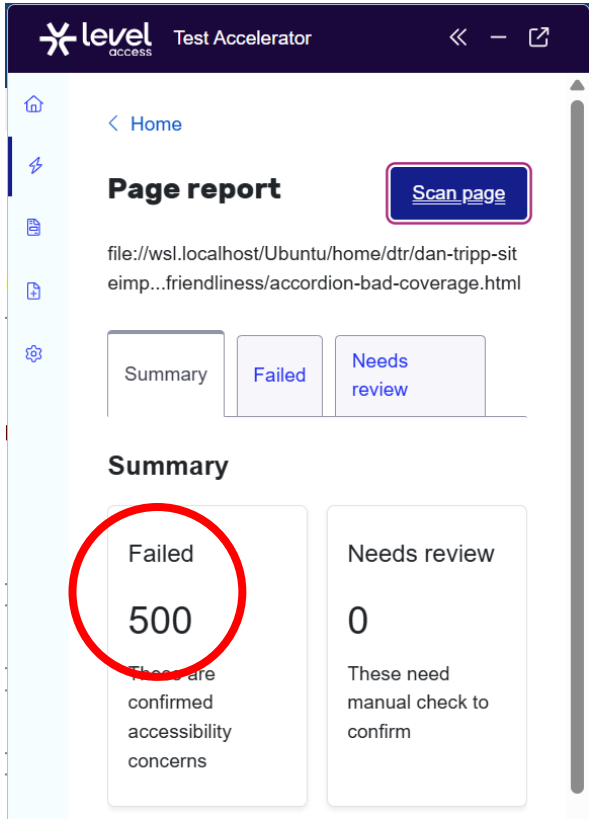
No issues detected for the chosen filter criteria. To see all issues, turn off focus view, reset the filters, and do not filter hidden issues.

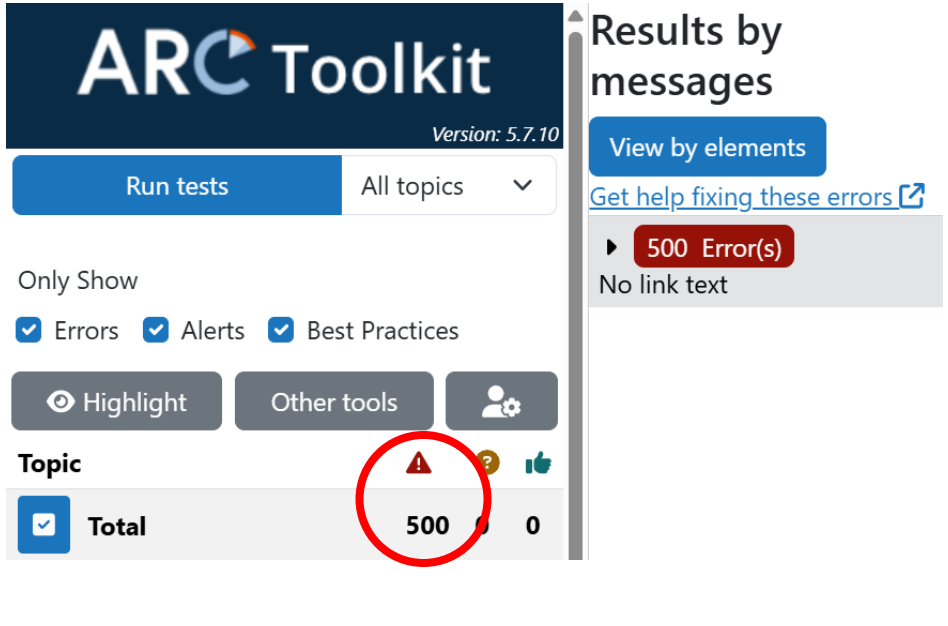


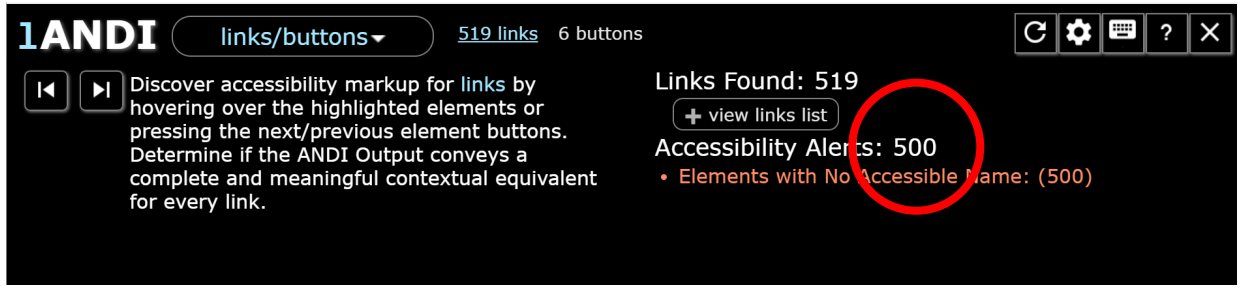
STORY #3 OF 8: ACCORDIONS

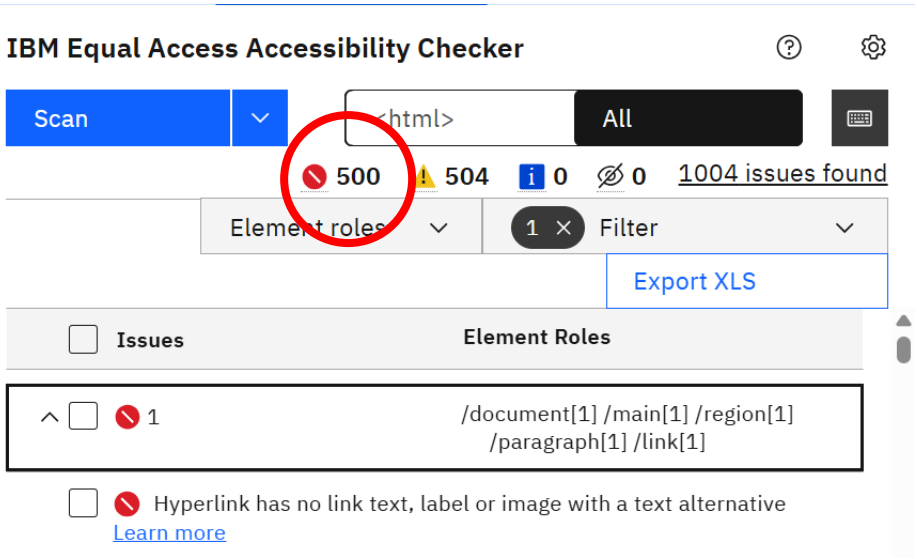
STORY #3 OF 8: ACCORDIONS

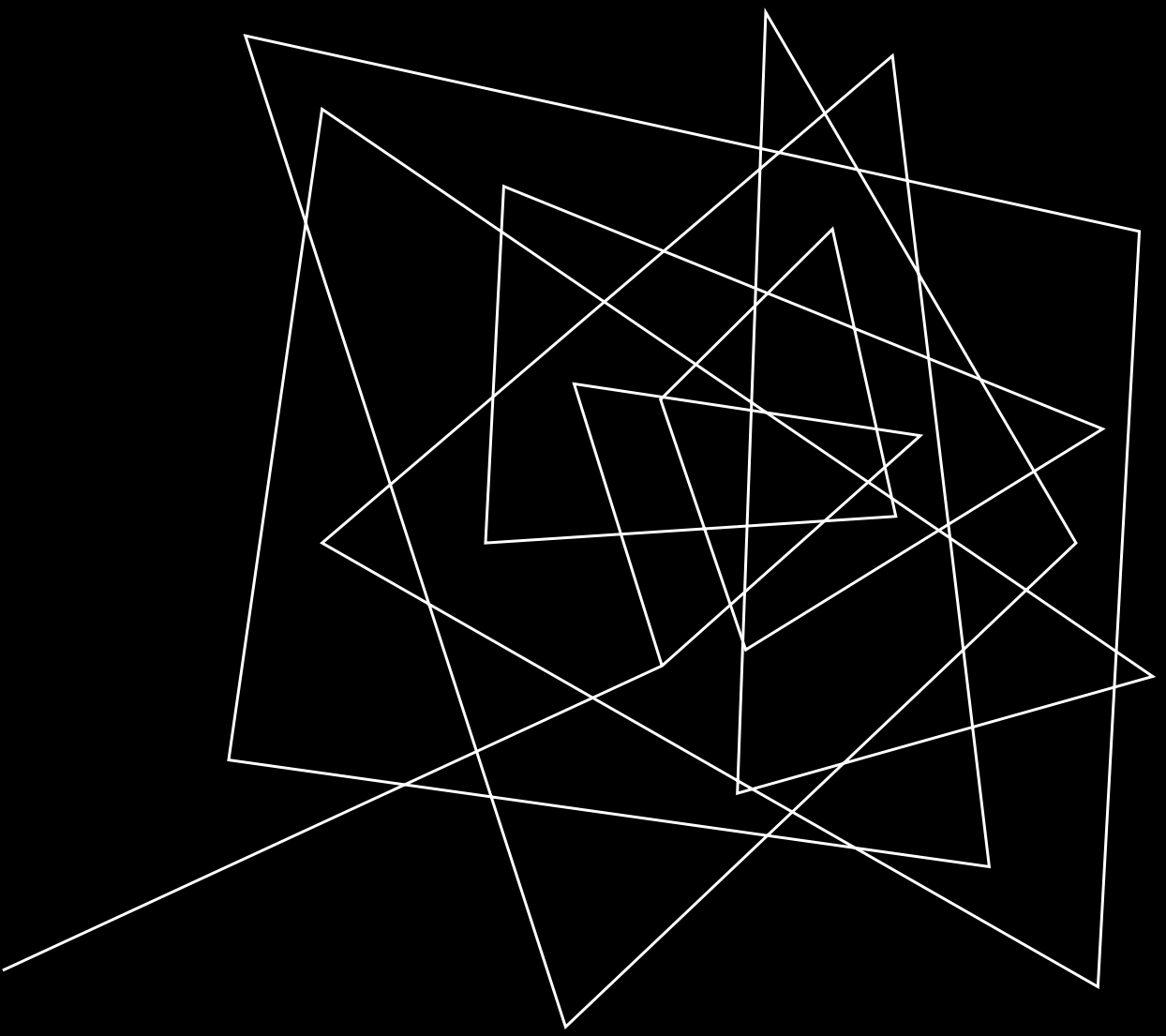






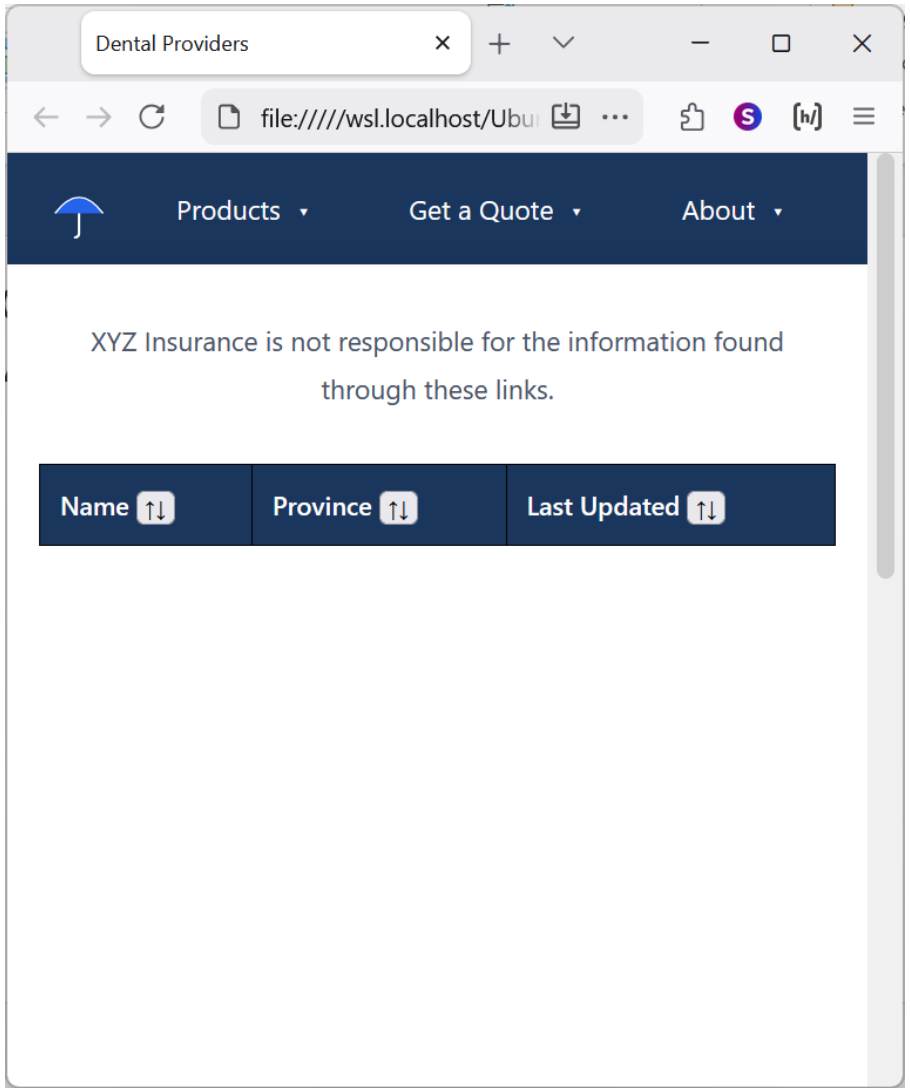




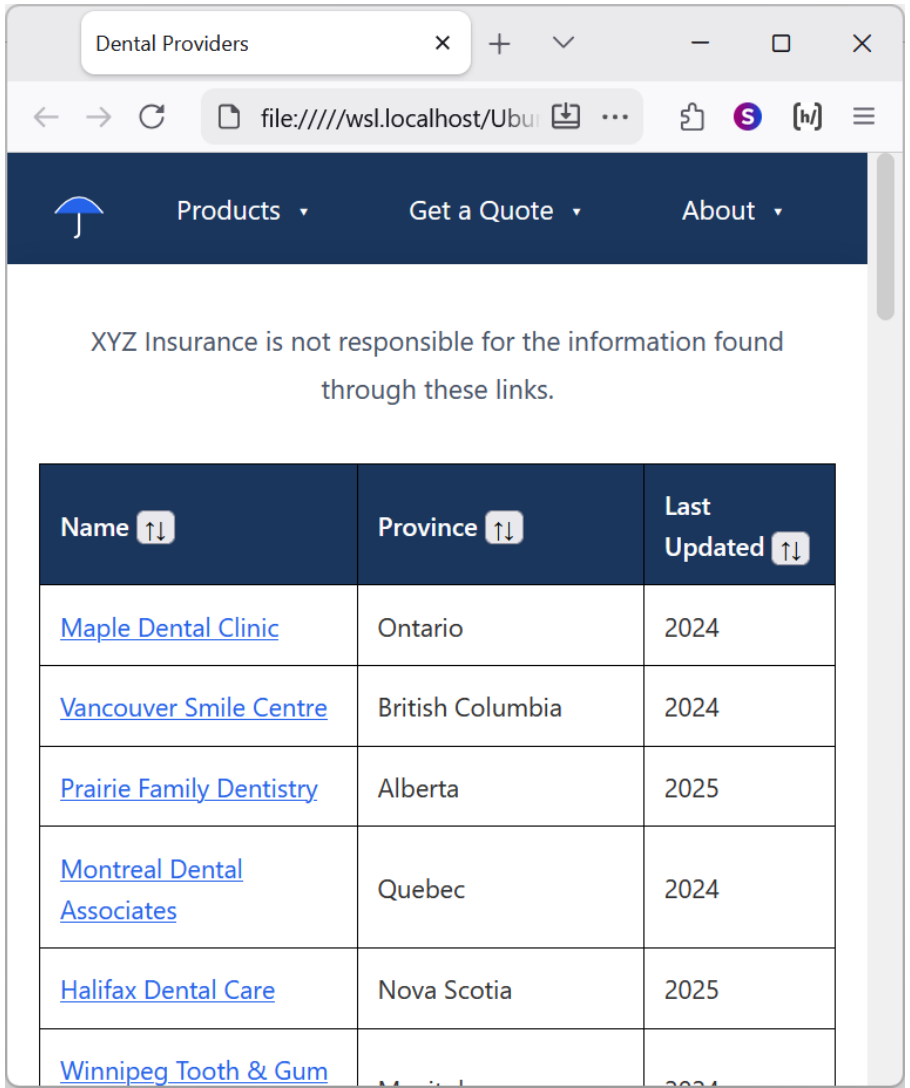


CUSTOMER STORY #4 OF 8: “CONTENT FROM API” AT A CANADIAN INSURANCE COMPANY

STORY #4 OF 8: “CONTENT FROM API”



A few ➔ seconds ➔ later



STORY #4 OF 8: “CONTENT FROM API”

- This site: empty. Other sites: a “placeholder”:

- Time is random.

STORY #4 OF 8: “CONTENT FROM API”

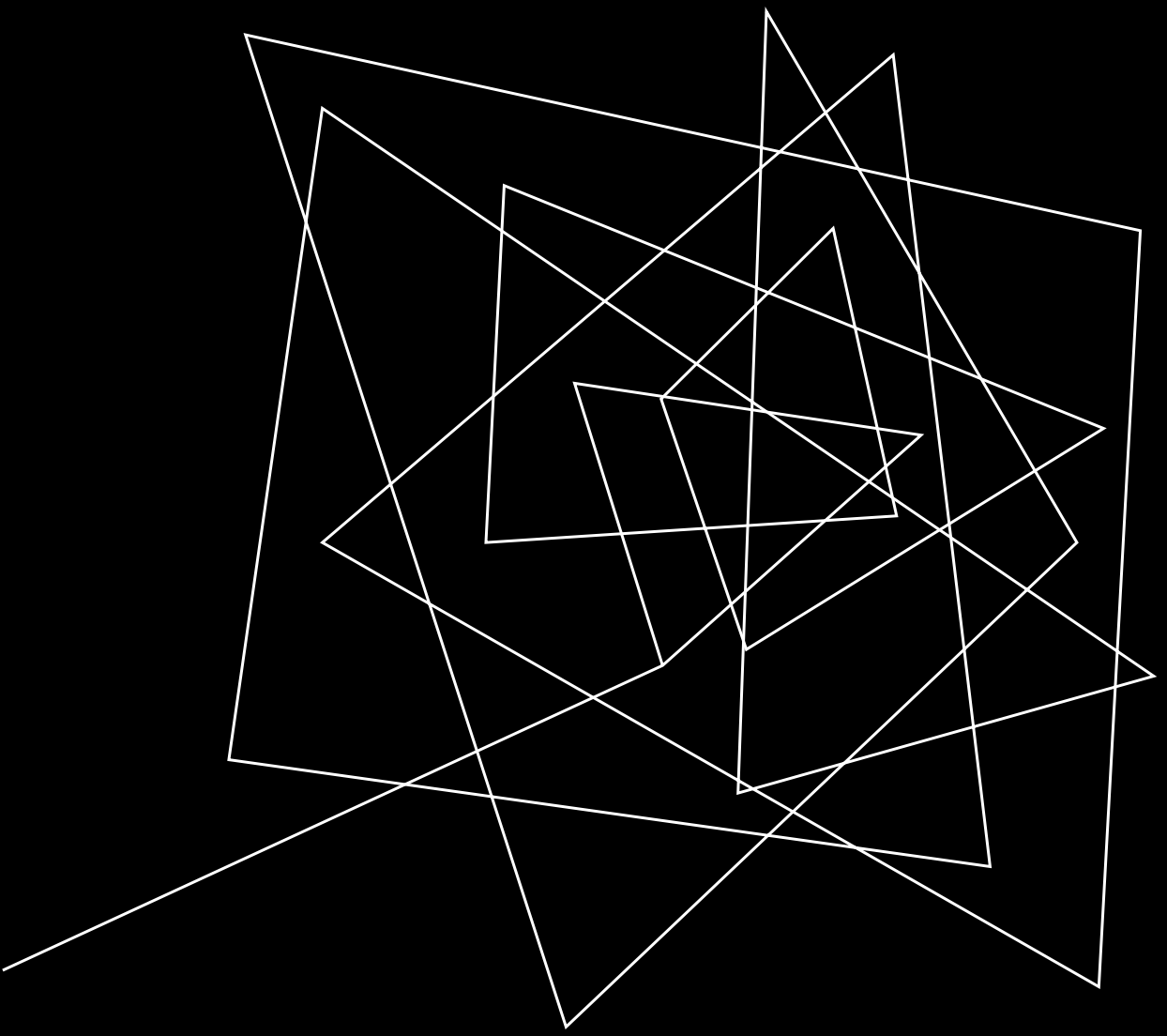
- Bad coverage
- Crawler-only problem
- A delay will fix it, right?
- 30 seconds x 20,000 pages = 7 days
- **No standard signal that the page has finished loading**
  - JS “load” event, puppeteer networkidle0 – sometimes.

STORY #4 OF 8: “CONTENT FROM API”

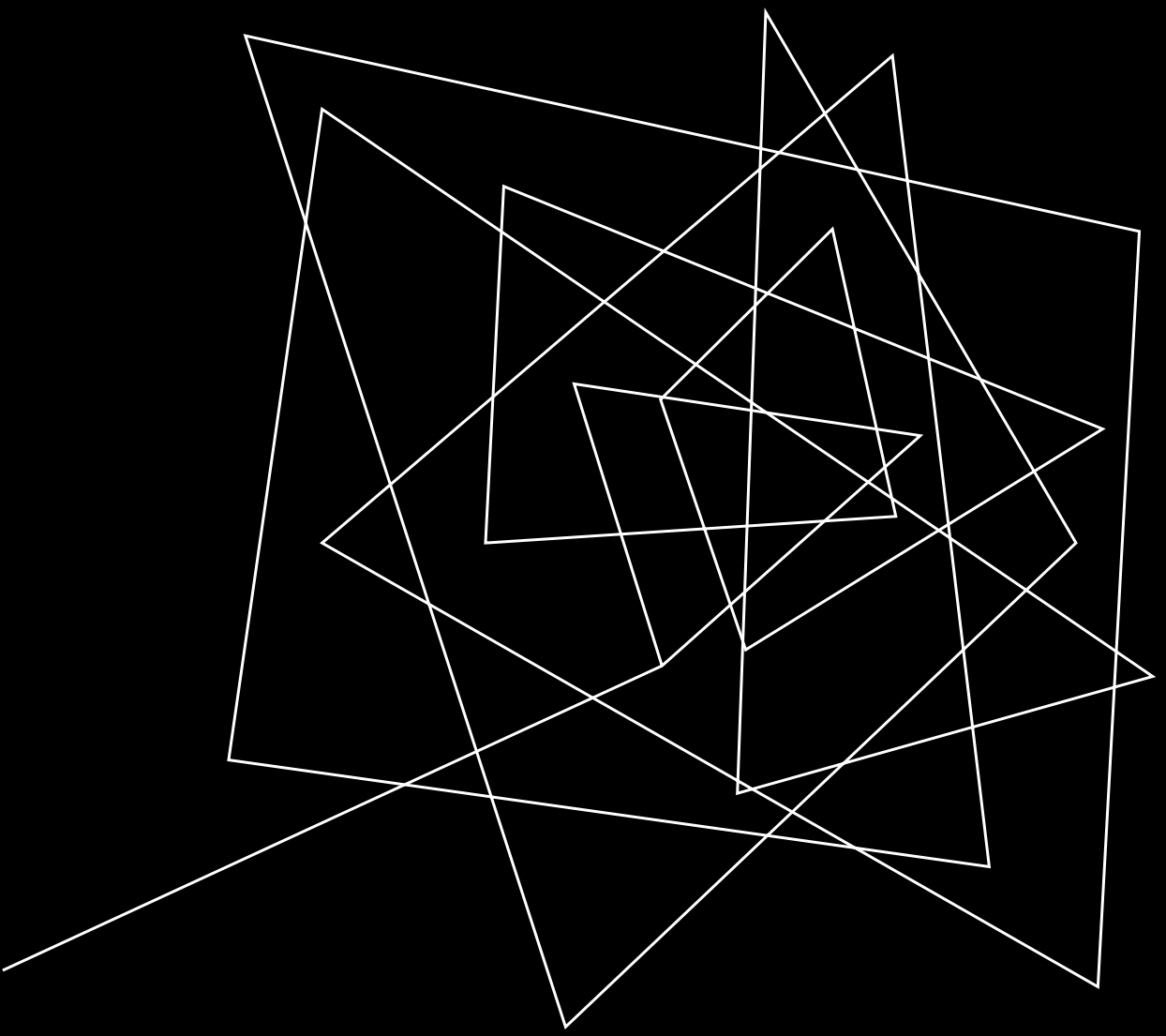
Options:

1. Customer lives with the bad coverage. ← **Worst**
2. We (the vendor) add a delay in the platform (we do this often).
3. Customer makes the *entire* page load fast by any means necessary.

   (Lighthouse performance metrics will tell you.)
4. Customer implements server-side rendering. ← **Best**

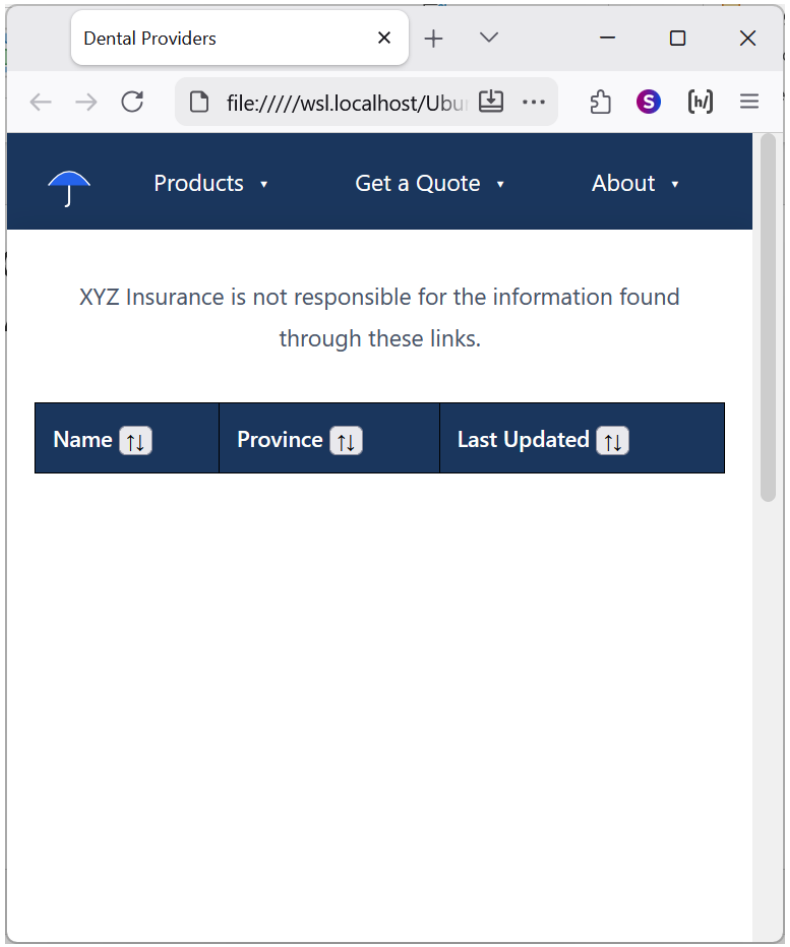


FRUSTRATION TYPE #2: NOISE

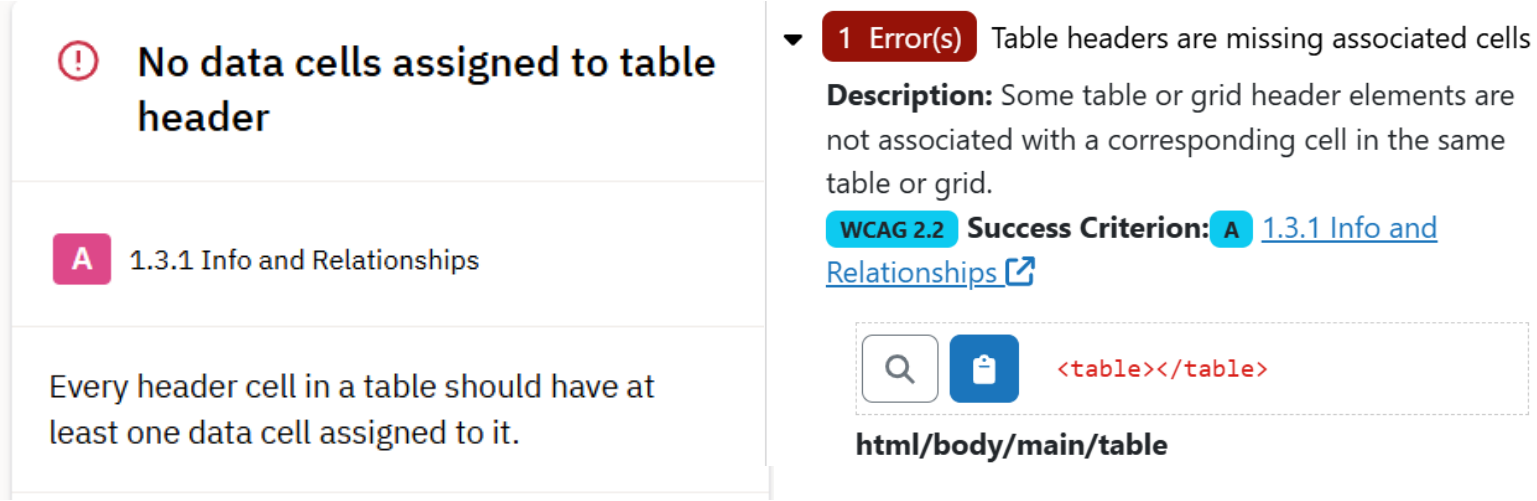


CUSTOMER STORY #5 OF 8: “CONTENT FROM API” AT A CANADIAN INSURANCE COMPANY *REVISITED*

STORY #5 OF 8: “CONTENT FROM API”



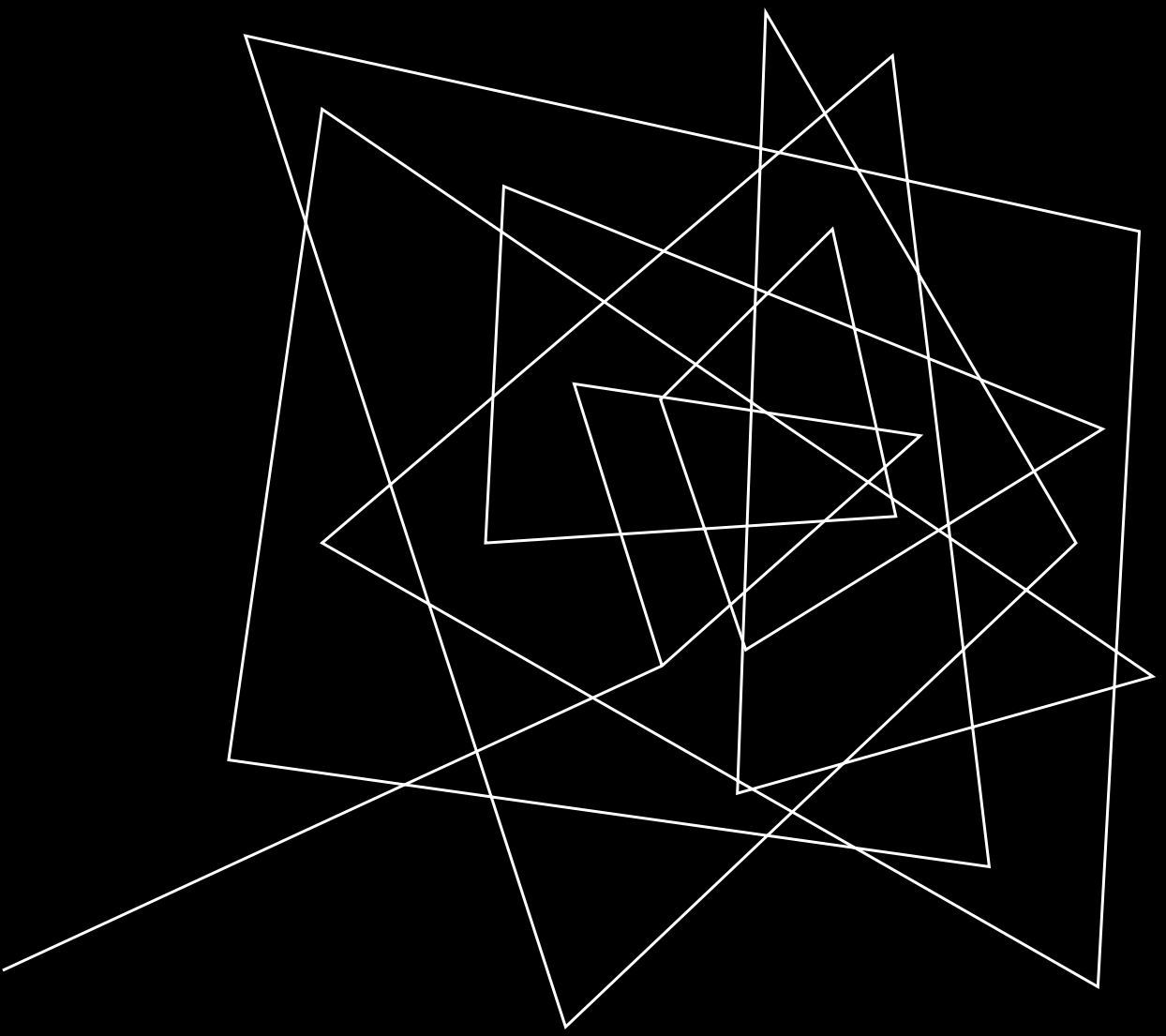
- Noise.
- Empty table failed by Siteimprove, ARC from TPGi.



STORY #5 OF 8: “CONTENT FROM API”

Option for noise:

- We (the vendor) hide the noise in the platform
  - (I did this, this time. Doesn’t help with bad coverage.)



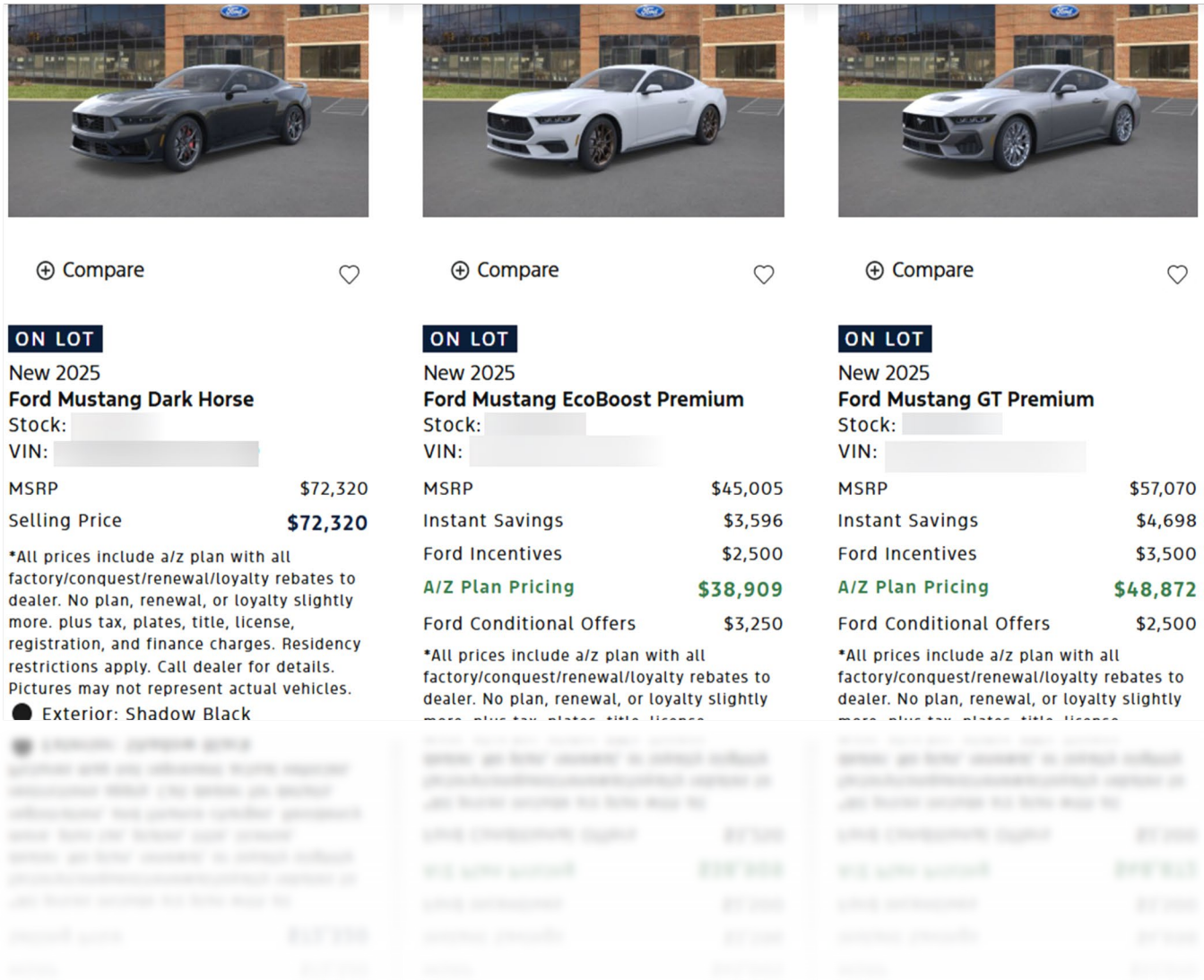
CUSTOMER STORY #6 OF 8: `GAMING THE CHECK` AT A CAR DEALERSHIP NETWORK

GAMING THE CHECK

- This is not what we mean by Automated Checker Friendliness
  - ACF is not just suppressing noise
- Stems from the desire to reduce noise.
  - Provides a false sense of security.

CUSTOMER STORY #6 OF 8: GAMING THE CHECK

- Consulting Call
- Customer complaint: Automated checker was flagging all the text on the page for “Not being in a landmark region”
- Customer interpreted this as noise
- Div Soup



CUSTOMER STORY #6 OF 8: GAMING THE CHECK

- The customer's solution:

```
<div role="region">Some text content</div>
```

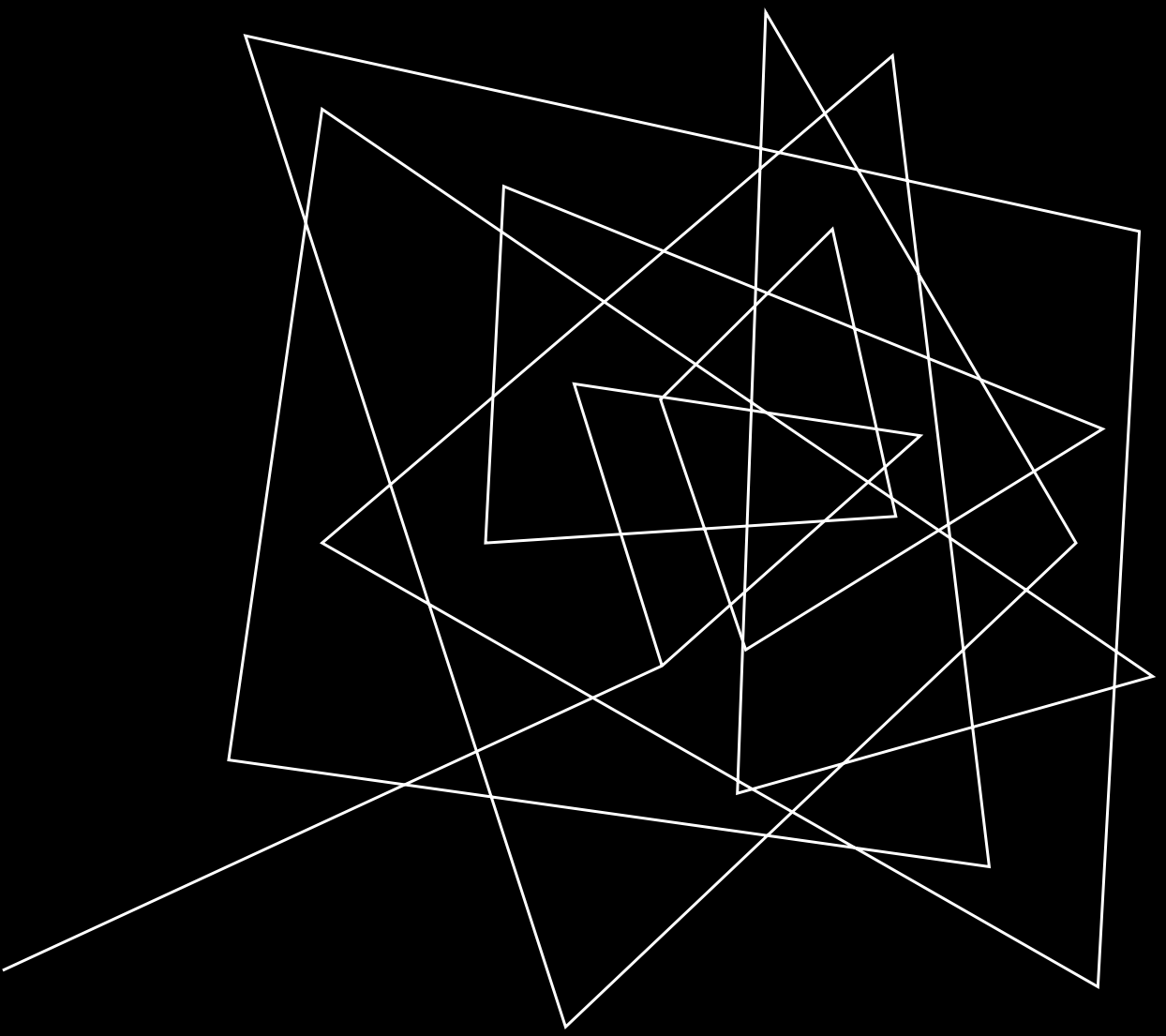
This satisfies the automated checkers

CUSTOMER STORY #6 OF 8: GAMING THE CHECK

- Made the user experience worse.
- Page is still missing structure and hierarchy of content.

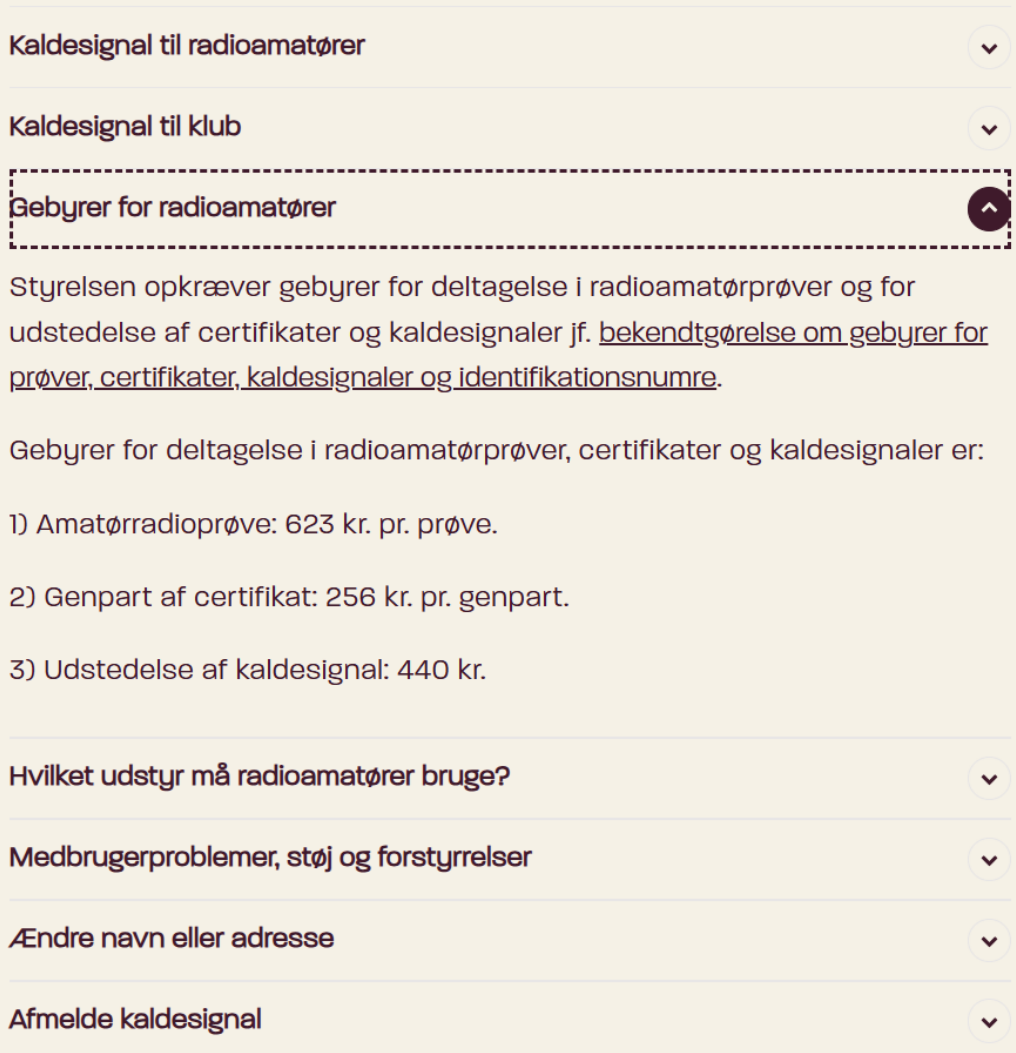
- Solution:

```
<main>…</main>
```



CUSTOMER STORY #7 0F 8: RULE ASSUMPTION VIOLATION AT A DANISH COMMUNICATIONS ORG

CUSTOMER STORY #7 OF 8: RULE ASSUMPTION VIOLATION



- False Positive: Elements with *aria-hidden* are not focusable.
  - This rule checks if interactive elements are hidden from AT users.
  - In this case, our platform was flagging links within the panel.

*Built differently than the accordion panels we saw earlier.

CUSTOMER STORY #7 OF 8: RULE ASSUMPTION VIOLATION

- Unique implementation
  - height:0px
  - Aria-hidden
  - Focus sentinel
- Customer argued: Automated checker is incorrectly flagging this, because their implementation would not allow focus to ever enter the collapsed panels

ACT RULES

Standardized set of testing rules that describe how to test the conformance of accessibility standards such, as ARIA and WCAG

- Helpful for both automated and manual testers
- Each rule describes how to test one aspect of a standard.
- Allows vendors to implement and build upon these rules

Cross Vendor Similarities

- ACT Rule - 6cfa84
  - SIA-R17: Elements with aria-hidden="true" are not focusable
  - axe aria-hidden-focus: ARIA hidden element must not be focusable or contain focusable elements

RULE ASSUMPTIONS

- Documented limitations of automated checkers
  - Violating assumptions will generate noise

STORY # 7 OF 8: RULE ASSUMPTION VIOLATION

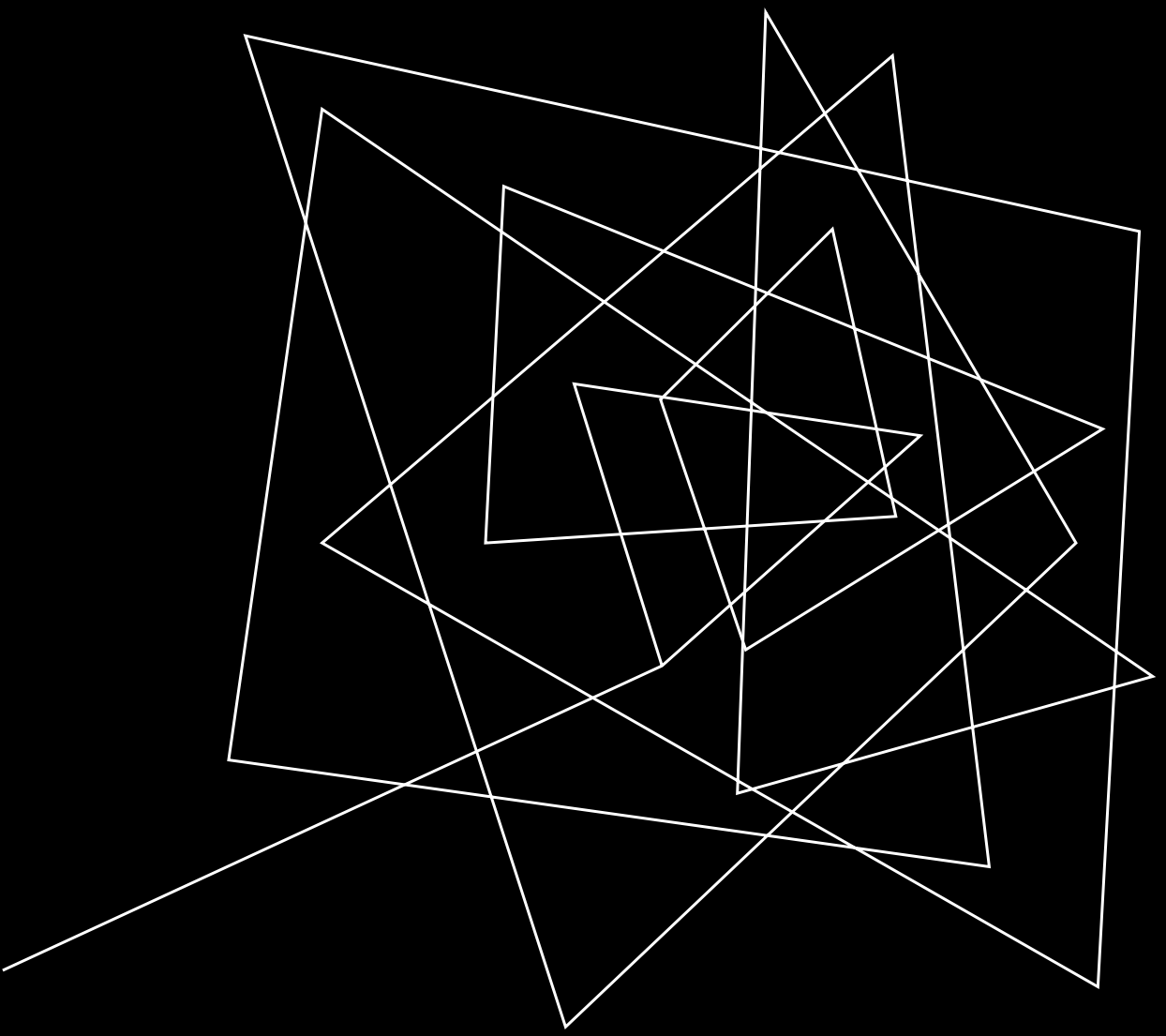
Rule Assumptions for SIA-R17

- Focusable elements do not redirect focus immediately upon receiving it.

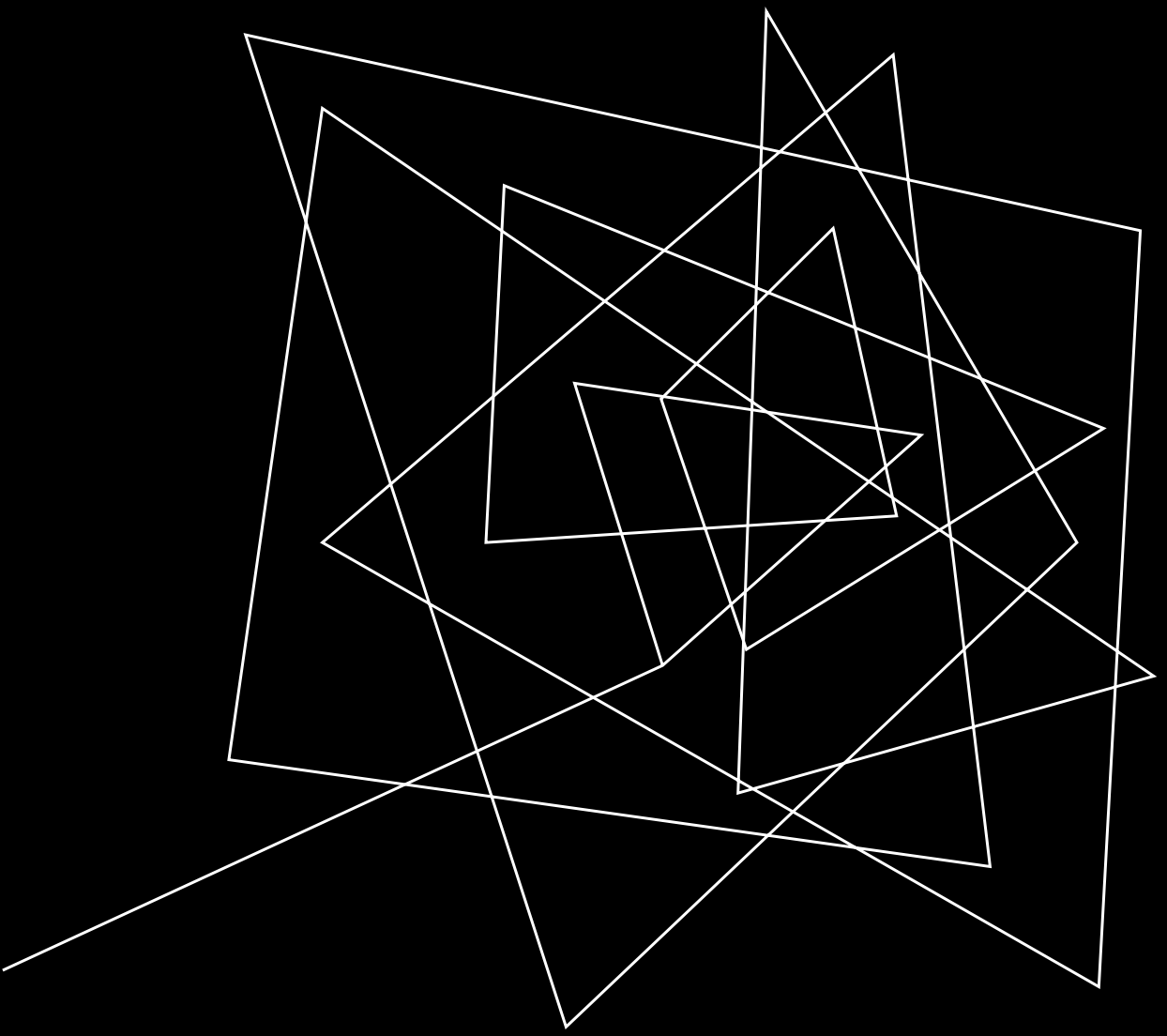
- The customer violated this second assumption with their focus sentinel.

CUSTOMER STORY #7 OF 8: RULE ASSUMPTION VIOLATION

- If your automated checker is generating a lot of noise, check your platform's documentation and ACT rules to ensure you're not violating any assumptions.
  - Documents are free and open source – All ACT Rules



FRUSTRATION TYPE #3: LOOKS LIKE NOISE (EVEN TO EXPERTS) BUT ISN’T



CUSTOMER STORY #8 OF 8: TYPEWRITER EFFECT AT A UNIVERSITY

TYPEWRITER EFFECT

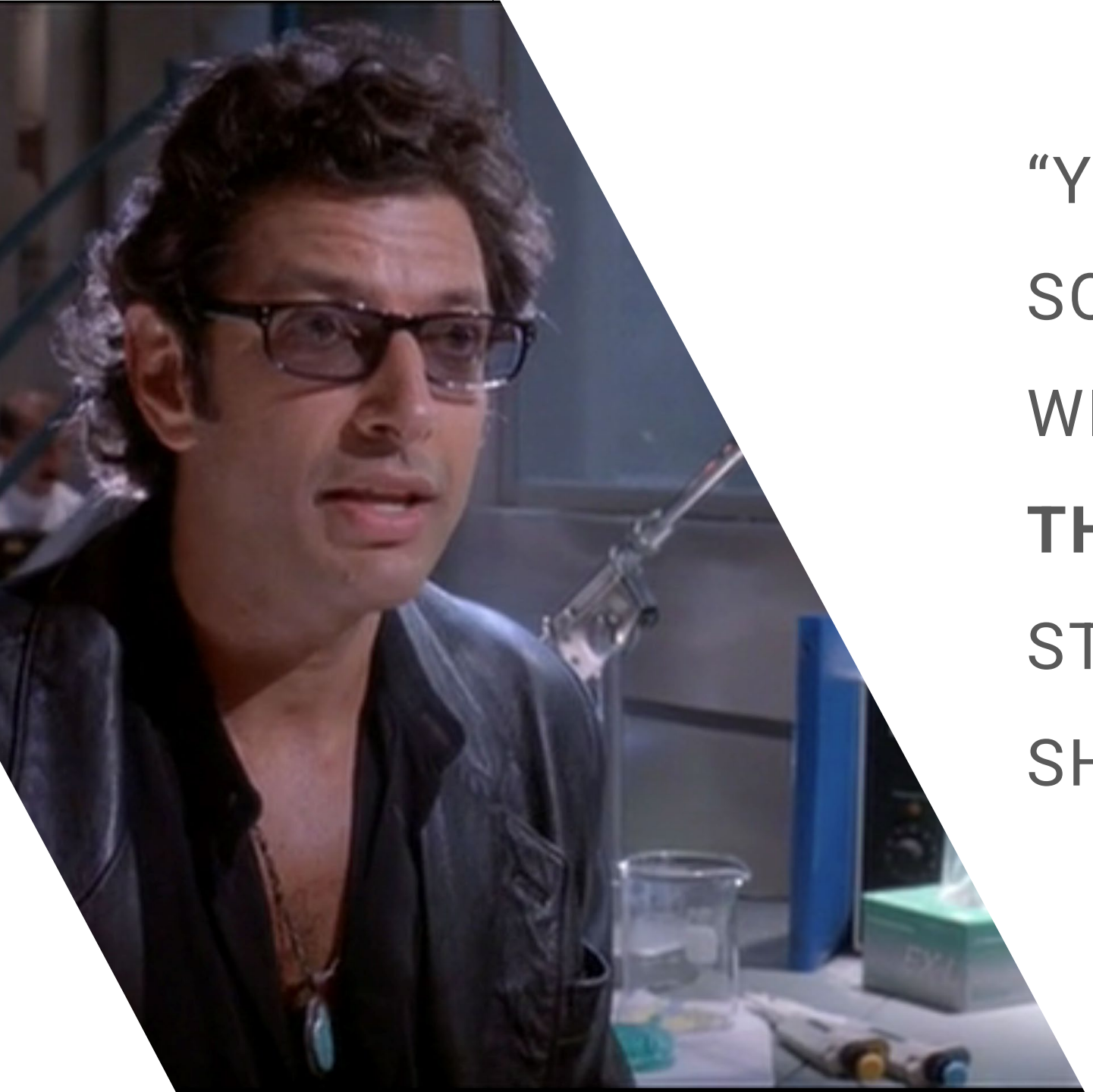
- Animated effect that looks like text is being typed on-screen.
- This isn’t just a visual animation. Content is actively being added and then removed from the DOM.
- Customer stated that the platform was intermittently flagging their page for an empty heading.

TYPEWRITER EFFECT

- Looks like noise, but isn’t.
  - Has true accessibility issues with AT
- Possible fixes:
  - **Best:** Don’t add/remove content to/from the DOM!
  - **Next Best:** Have a static heading that is placed offscreen and apply aria-hidden to animated content

MORE IMPORTANTLY

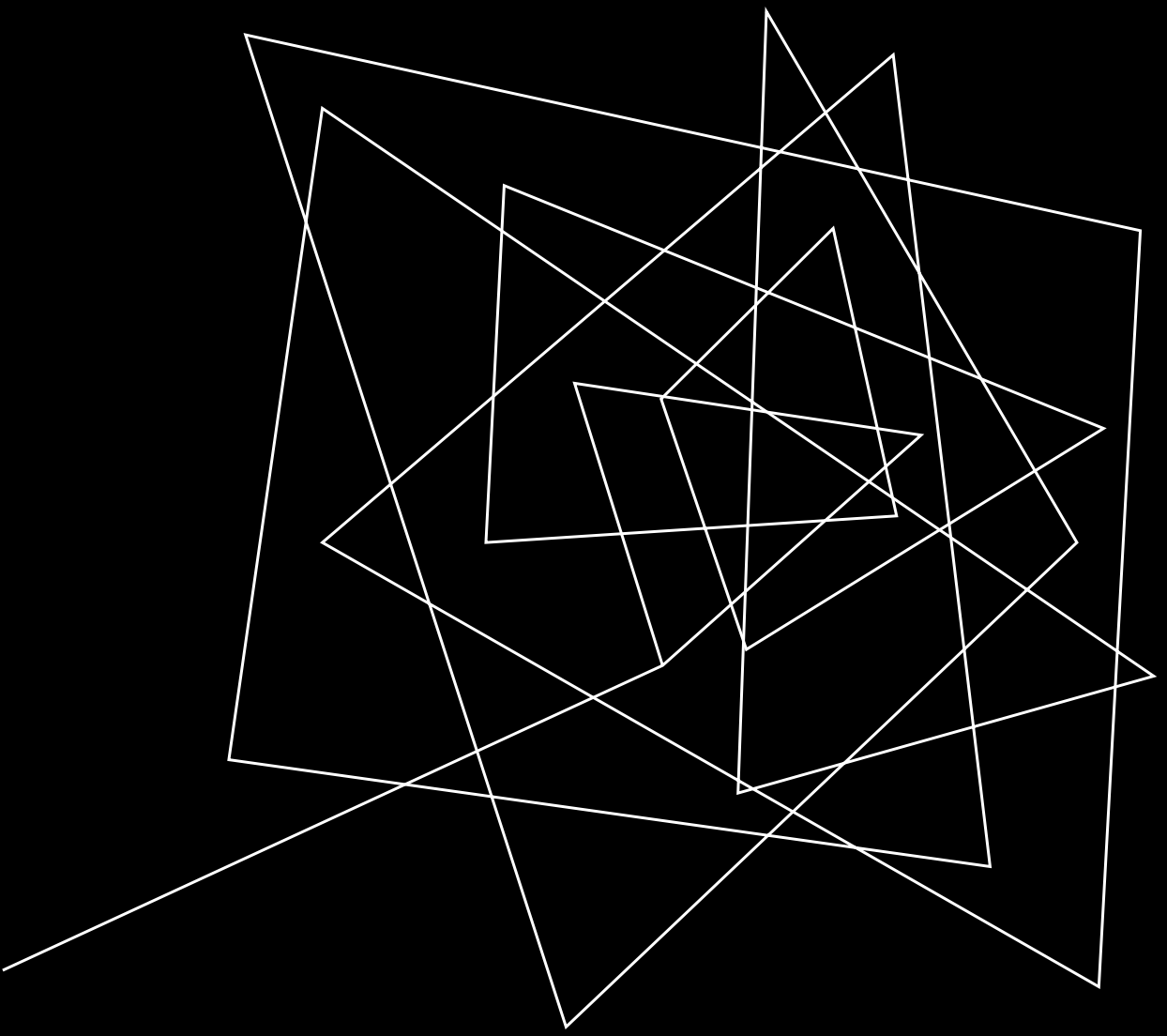
- It’s not just this typewriter effect!
- This is often seen when chasing the latest web design trends.
  - Rarely do these trends account for accessibility.
  - The further you stray from the semantic markup and accessible patterns, the more at risk you become for Automated Checker Unfriendliness.



"YOUR SCIENTISTS WERE SO PREOCCUPIED WITH **WHETHER OR NOT THEY COULD**, THEY DIDN'T STOP TO THINK IF THEY SHOULD."

- IAN MALCOM

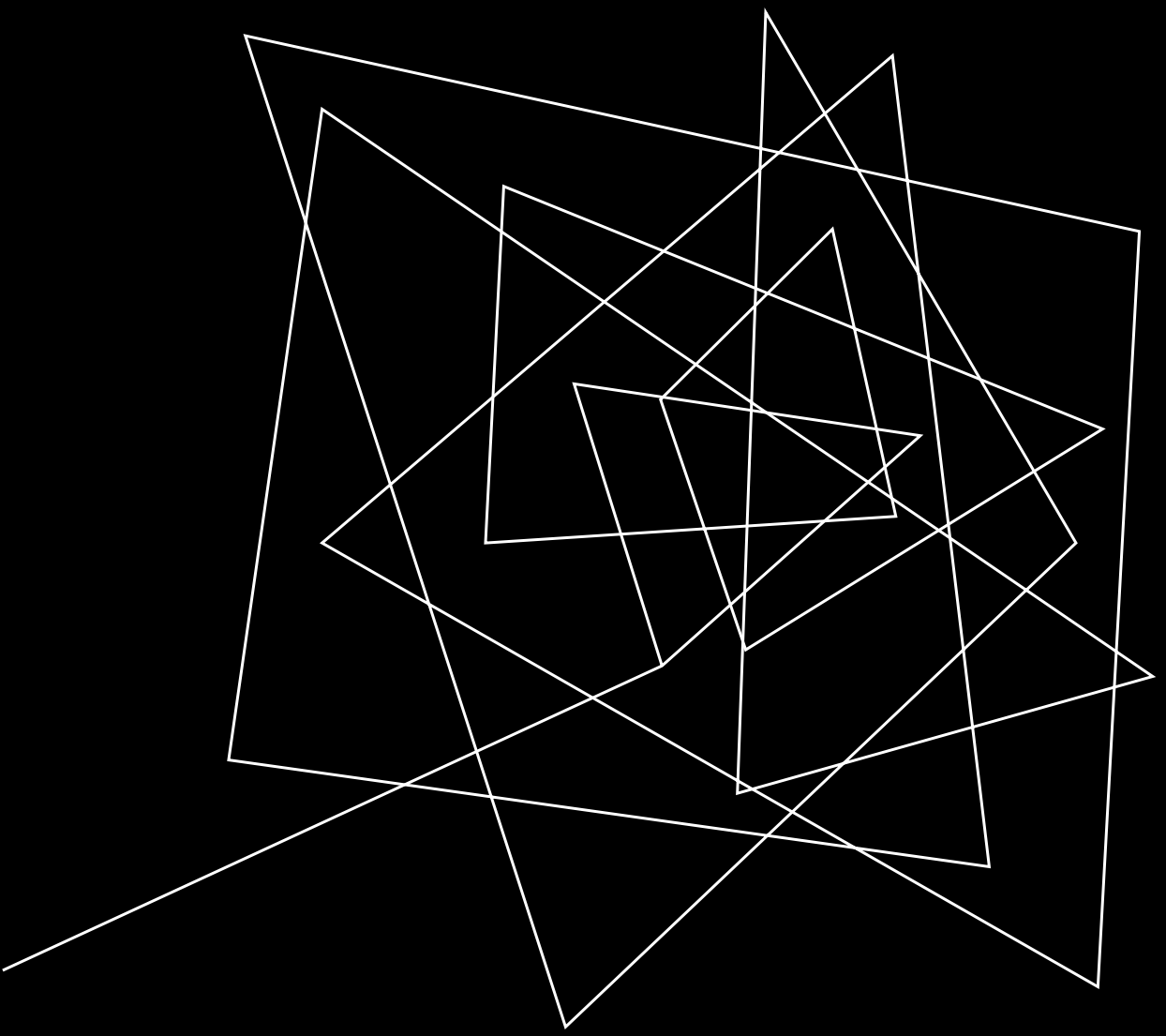
Favor predictability over individuality



CONCLUSION

CONCLUSION

- For every one of our eight customer stories: we’ve seen the same thing at tens if not hundreds of other customers.
- We cut approx. 14 stories from this talk, due to the 40 minute talk length.
- Again: why does Automated Checker Friendliness matter?
  - Because increased coverage + less noise ➔ time savings ➔ more efficient accessibility program ➔ better accessibility
- **Don't be like Homer Simpson. Be like Ian Malcolm instead**



GET IN TOUCH

jgo@siteimprove.com
https://www.linkedin.com/in/jeremy-gonzales1331/

dtr@siteimprove.com
https://ca.Linkedin.Com/in/dan-tripp